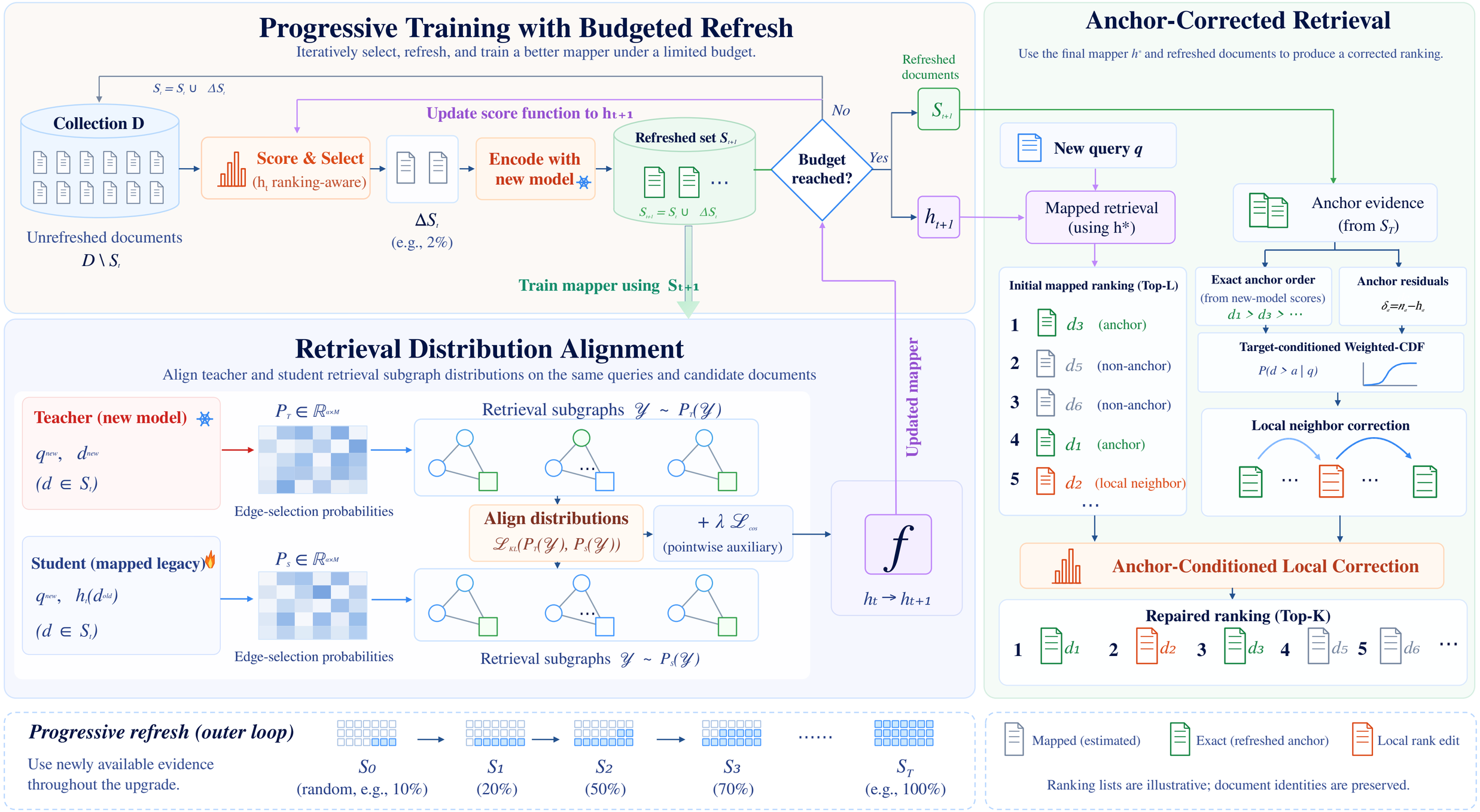

Progressive Training with Budgeted Refresh
Iteratively select, refresh, and train a better mapper under a limited budget.
$S_t = S_t \cup \Delta S_t$
Collection D
Unrefreshed documents
$D \setminus S_t$
Score & Select
(h_t ranking-aware)
Update score function to h_{t+1}
ΔS_t
(e.g., 2%)
Encode with new model
Refreshed set S_{t+1}
$S_{t+1} = S_t \cup \Delta S_t$
Budget reached?
No
Yes
Refreshed documents
S_{t+1}
h_{t+1}
Train mapper using S_{t+1}
Retrieval Distribution Alignment
Align teacher and student retrieval subgraph distributions on the same queries and candidate documents
Teacher (new model)
q^{new}, d^{new}
$(d \in S_t)$
$P_T \in \mathbb{R}^{\alpha \times M}$
Edge-selection probabilities
Retrieval subgraphs $\mathcal{Y} \sim P_T(\mathcal{Y})$
Align distributions
$\mathcal{L}_{KL}(P_T(\mathcal{Y}), P_S(\mathcal{Y}))$
$+ \lambda \mathcal{L}_{cos}$
(pointwise auxiliary)
Student (mapped legacy)
q^{new}, $h_t(d^{old})$
$(d \in S_t)$
$P_S \in \mathbb{R}^{\alpha \times M}$
Edge-selection probabilities
Retrieval subgraphs $\mathcal{Y} \sim P_S(\mathcal{Y})$
Updated mapper
f
$h_t \rightarrow h_{t+1}$
Anchor-Corrected Retrieval
Use the final mapper h^* and refreshed documents to produce a corrected ranking.
New query q
Mapped retrieval (using h*)
Anchor evidence (from S_T)
Initial mapped ranking (Top-L)
1 d_3 (anchor)
2 d_5 (non-anchor)
3 d_6 (non-anchor)
4 d_1 (anchor)
5 d_2 (local neighbor)
Exact anchor order
(from new-model scores)
$d_1 > d_3 > \cdots$
Anchor residuals
$\delta_a = n_a - h_a$
Target-conditioned Weighted-CDF
$P(d > a \mid q)$
Local neighbor correction
Anchor-Conditioned Local Correction
Repaired ranking (Top-K)
1 d_1 2 d_2 3 d_3 4 d_5 5 d_6
Progressive refresh (outer loop)
Use newly available evidence throughout the upgrade.
S_0 (random, e.g., 10%)
S_1 (20%)
S_2 (50%)
S_3 (70%)
S_T (e.g., 100%)
Mapped (estimated)
Exact (refreshed anchor)
Local rank edit
Ranking lists are illustrative; document identities are preserved.